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Bitcoin Usage: Study on Bitcoin Usage Around the World 2020

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Bitcoin usage has grown from zero to hundreds of billions of dollars in just over a decade. However, the research and studies of bitcoin usage fall short. This is mainly due to the nature of bitcoin, but also the myths of the early days. In this study, we go through the user data from LocalBitcoins, the largest peer-to-peer bitcoin marketplace in the world to study the bitcoin usage and needs. By analyzing and classifying major use cases for bitcoin through surveys and statistical analysis, this study will bring a better understanding of the different use cases for bitcoin and the role they play in bitcoin adoption. This research shows that the scope of bitcoin usage is noticeably more diverse and not merely limited to illegal usage or speculation.

Keywords: Bitcoin; usage; adoption.

1. Introduction

There is a wide misconception of Bitcoin[1] and cryptocurrencies in general as they are often thought to be used only for speculative trading. This is likely due to a limited understanding of bitcoin usage worldwide, generally causing many professionals to view cryptocurrencies as a speculative zero-sum game

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[1]The word "Bitcoin" appears both capitalized and with a lowercase first letter in this study. "Bitcoin" with a capital B refers to Bitcoin as the protocol, network, or idea, whereas "bitcoin" refers to the currency or the unit of account. https://bitcoin.org/en/vocabulary#address.

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only, and thus viewing them as a bubble (La Roche, 2020; Krugman, 2018). Bitcoin has been said not to "serve any socially useful function" and is "mainly used for illegal financing" (Hochstein, 2017; Robertson, 2021).

The aim of this study is to shed light on this issue consequently providing a better understanding of various existing use cases as well as the factors affecting bitcoin adoption.

LocalBitcoins has a unique position as it is the largest and most widely used peer-to-peer bitcoin exchange platform in the world with over 6 million registered users. This provides a unique opportunity to better understand bitcoin usage, as well as the shortcomings of the existing traditional financial infrastructure.

This study analyzes and classifies major use cases for Bitcoin. The magnitude of each use case is also evaluated, with the aim of truly understanding what role they play in the adoption phase. Finally, each of these is evaluated case by case providing an understanding of the reasons the use cases exist.

This study is conducted through surveys and statistical analysis methods. Surveys and qualitative analysis have been carried out to understand the different categories of needs. Finally, a more in-depth survey using quantitative analysis is carried out to verify findings.

This paper is formatted as follows; Sec. 2 presents related work. Section 3 presents the classification of usage categories and the proportion they play in the adoption. Section 4 presents an analysis of each category. Section 5 presents a link between the main socio-economic drivers and bitcoin use cases. Section 6 presents the conclusion of this study.

2. Related Works

Studying bitcoin usage is essential for understanding the future of decentralized cryptocurrency and its possible socioeconomic impact.

Despite the fact that blockchain is a relatively new technology, a significant number of studies have already been conducted with the aim of understanding its function and development (Bartoletti *et al.*, 2019; Antonopoulos, 2017). Another important line of research has been dedicated to the understanding of legal matters such as regulatory policy (Capaccioli, 2020; Fletcher *et al.*, 2021).

The most relevant studies can be grouped with relation to the scope of the collected data. The first group includes bitcoin usage case studies by country.

The most recent work was carried out by Henry *et al.* focusing its attention on Bitcoin users in Canada (Henry *et al.*, 2019). The results of their research

included data from a survey called the Bitcoin Omnibus Survey (BTCOS), which was conducted by the Bank of Canada in 2016 and 2017. Such an approach allows researchers both to observe the increase in adoption rate as well as understand the motivational shift of bitcoin holders.

Similar methods were employed by Tsanidis *et al.*, who examined Bitcoin users in Greece (Tsanidis *et al.*, 2015), namely, through their survey of 130 random people, which demonstrated a skeptical attitude towards Bitcoin amongst the Greek population.

The second group is related to worldwide bitcoin adoption and usage. In their study, Athey *et al.* (2016) combine an analysis of the Public Ledger with a review of relevant literature in order to define the main bitcoin use cases around the world.

3. Method

This study employs descriptive statistics methods. All together 14 surveys were sent out to 1.2 million LocalBitcoins users globally and over 66,000 answers were recorded from these surveys. The process started with an open-ended question aiming to gain an understanding of the reasons behind people using bitcoin. The first survey was sent to over 100,000 LocalBitcoins users worldwide. Over 7,000 replies were received, which neatly represented the global user base of LocalBitcoins. From this survey, the first list of potential usage categories was then compiled.

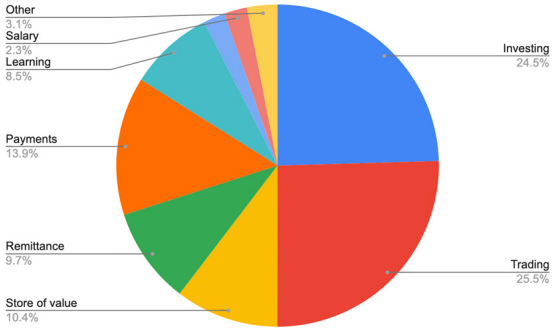
To test and refine these categories a set of surveys were conducted with predefined options. An open questionnaire was used to manually form the options. Following this, surveys were sent to a randomized set of users with the option of choosing a predefined purpose or the option to answer, "other purpose". Textual descriptions and other information were collected to further clarify the answers.

In the first surveys the "other" option contained answers that were not part of the main categories, thus the set of categories was refined in each round. The process was continued for 12 surveys receiving over 4,000 replies in total. After this there was not a significant volume of answers in the " other" option.

Finally, the last survey was sent out to almost one million LocalBitcoins users globally and localized into 7 languages — English, Spanish, French, Chinese, Portuguese, Russian and Italian. Over 55,000 answers were recorded.

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4. High Level View on Usage Categories



The following categories were identified for bitcoin usage globally. The categories which were also presented to participants of the survey include:

(1) Investing — I buy bitcoin and hold it for a long-term price increase.
(2) Trading — I frequently buy and sell bitcoin to make profit.
(3) Store of value — I buy bitcoin to protect myself from local risks, such as the economy, currency and politics.
(4) Remittance — I send or receive bitcoins across borders.
(5) Payments — I buy goods and services and pay with bitcoins.
(6) Learning — I want to learn and better understand what Bitcoin is all about.
(7) Gambling — I use bitcoin to gamble online.
(8) Salary — I use bitcoin to pay for work and services by others (such as freelancers or IT workers).

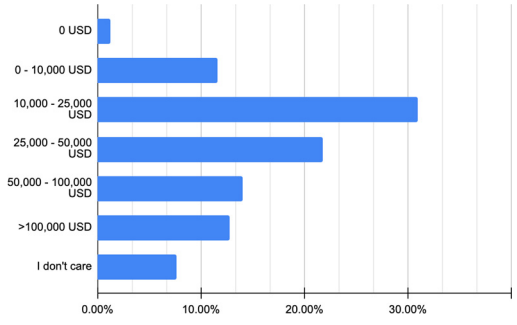
Next, we evaluate this classification category by category.

4.1. *Investing*

Investing is defined in this study as acquiring and holding bitcoins for an extended period of time with the hope of a future price increase. 13,212 respondents (24.5% of total) stated that they buy bitcoin for investment purposes.

What was the estimated price increase? In each survey, users were asked what they expected the price of bitcoin to be in five years. At the time of these surveys, bitcoin was slightly under 10,000 USD, which provides a baseline of what kind of increase users using bitcoin for the purpose of investing, believed they would see within five years.

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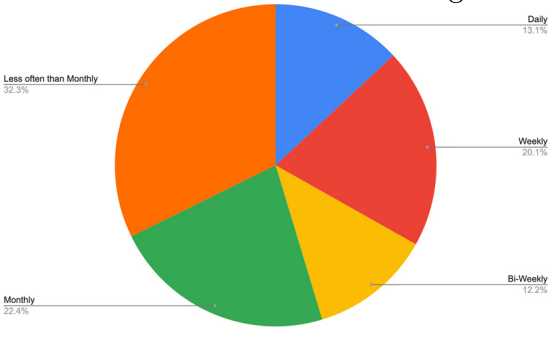
As can be seen, over 50% of users were expecting the price of bitcoin to be over 25,000 USD within five years and almost one-third, over 50,000 USD. Expectations like these naturally support one's willingness to invest and hold bitcoin in the long term. However, now, a few months after conducting these surveys, the price of bitcoin has already increased to over 50,000 USD, hence the described price expectations above would surely be different today.

4.2. *Trading*

Trading is defined in this study, as the frequent buying and selling of bitcoin to make profits from short-term swings. For a trader, the long-term price development is of secondary importance compared to volatility. 13,717 respondents (25.5% of total) stated that they buy and sell bitcoin for trading purposes.

Results demonstrated that trading was a very prominent use case especially in Africa, where people set out to earn extra income through bitcoin trading. The qualitative research conducted — interviews with LocalBitcoins customers in African countries — confirmed prior quantitative findings.

Additionally, we asked the responders whose primary use case was trading about their trading habits and frequency of trading, and found out that over half of the traders do at least one trade a month, while one-third trade weekly or even daily. The chart below visualizes the frequency of trading by LocalBitcoins traders (those whose main use case is trading for extra income).



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4.3. *Store of value*

Store of value use case was defined in this study by buying bitcoin to protect oneself from local risks, such as risks related to economy, currency and politics. 5,570 respondents (10.35% of total) stated that they use bitcoin to store the value of their wealth.

Given that there are a lot of countries globally with an unstable government or economics — at least to a certain degree, with some countries experiencing very high inflation — some even hyperinflation — or some kind of capital controls and government-dictated exchange rates that are far from street rates — bitcoin was used by a large portion of LocalBitcoins users for storing the value of their hard-earned money.

This is natural, since in a country whose currency has a double-digit annual inflation, it makes no sense to store wealth in local currency, and at the same time there are no global currencies available with decent exchange rates — such as EUR or USD.

Additionally, unstable governments with unpredictable politics incentivize people to turn to bitcoin for a store of value that cannot be confiscated.

As an example, in Argentina and Venezuela this use case was by far the biggest use case, with 23% of respondents saying that they buy bitcoin for reasons around store of value.

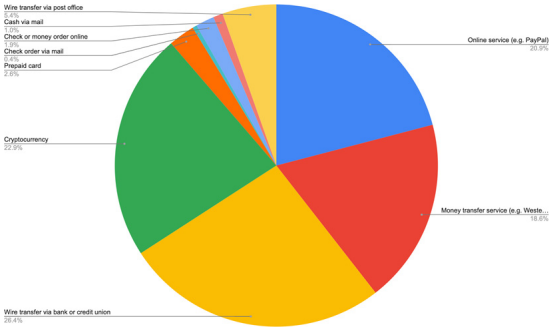
4.4. *Remittance*

Remittance was defined in this study by using bitcoin to send and receive money across borders. 5,206 respondents (9.7% of total) said that they use bitcoin for remittance.

While almost 10% of LocalBitcoins users use bitcoin for cross-border money transmissions, they had also tried many other money transmission services. However, of all available options for transferring money, bitcoin was the second most popular right after traditional wire transfer through a bank or credit union.

The following chart displays the most frequently used methods to send and receive money across borders among LocalBitcoins users, whose primary use case for Bitcoin is remittance.

Additionally, other studies we have done recently on this topic show that the reason to use bitcoin for remittance might be the high costs of traditional means for transferring money. For example, transferring money from the USA or Europe to Africa with traditional means comes with a 5%–10% transfer cost, while international money transfer through bitcoin and LocalBitcoins can achieve the same with only a 1%–3% transfer fee.



4.5. *Payments*

Payments use case was defined in this study as buying goods and services with bitcoin.

Interestingly it was discovered that almost 35% of the LocalBitcoins user base had used bitcoin to buy goods and services, and 7,479 respondents (13.9% of total) of our users identified payments as their main purpose for using bitcoins.

A similar method was used for identifying bitcoin usage categories to understand the categories of the goods and services our users buy with bitcoin. That is a topic to be opened in a whole new study, however the categories are briefly mentioned here:

(1) Web/e-things, such as e-books, online training, subscriptions to entertainment services, software packages, games, gift cards and conference tickets.
(2) Electronics: phones, accessories, laptops, watches, gaming consoles, camera lenses, drones.
(3) Equipment: fashion accessories, cars, mountain bike parts, motorcycles, luxury goods, handbags.
(4) Bitcoin-related items: mining equipment, mining plans, bitcoin vaults.
(5) Blackmarket items.

Some users do also seem to be using bitcoin for purchasing illicit goods from dark markets. However, in accordance with Chainanalysis' report the percentage of illegal usage of all bitcoin transactions, such as using bitcoin for purchases from darknet markets has been at less than 1% in the last few years.

Due to the wide misconception that bitcoin is mainly used for illicit purposes, this is certainly an important and interesting topic to explore in future studies.

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4.6. *Learning*

The use case of Learning — "I want to learn and better understand what Bitcoin is all about."

4,594 respondents (8.5% of total) said that they buy bitcoin to learn more about it and how it works.

Learning is one of the primary use cases for using LocalBitcoins and bitcoin. Bitcoin itself is raising a lot of interest globally, however many users only want to buy modest amounts for use, in order to better understand what it is about and how it works.

It does seem that bitcoin is still in the early adoption phase.

4.7. *Gambling*

Gambling was defined in this study by buying bitcoin to gamble online. 1,181 respondents (2.2% of total) said that they use bitcoin to gamble online.

Gambling was one of the first major use cases for Bitcoin in its early years, and bitcoin is still an easy way to access online casinos worldwide.

Furthermore, gambling is still a relevant use case for bitcoin users, as more than 2% of users surveyed by LocalBitcoins use bitcoin primarily for gambling and for betting credits on global online casinos.

4.8. *Salary*

Salary was defined in this study as using bitcoin to pay for work and services carried out by others (such as freelancers or IT workers). 1,229 respondents (2.3% of total) said that they use bitcoin to pay for work and services.

PayPal goes a long way when it comes to global payments. However, we found out that bitcoin is increasingly used for paying, for example, for freelancing services bought across borders. Reasons for this are multiple, one being that PayPal is either banned or limited in many countries (for example, Turkey and many African countries), whereas bitcoin works everywhere. The driving forces are likely the same as in remittance.

LocalBitcoins users were asked which freelancing services they paid for with bitcoin, and we got a wide array of replies from writing services, design services and consultation to advertising, search engine optimization (SEO) services and email lead lists for email marketing purposes.

From these answers, one may deduct the conclusion that if one is working as an expert in the intellectual property industry, where no tangible items are delivered, bitcoin is actually very well-suited to buying, selling and paying for those services, especially when it comes to cross-border/international transactions.

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4.9. *Illicit usage*

It should be noted that some Bitcoin circulation happens in a grey area of illegal or questionable activities such as drug purchase, money laundering, and terrorism financing. The authors of the article do understand that this fact can create some bias outcomes as a result of omitting to provide complete information by the respondents. However, we cannot expect the users of the platform to reveal their use of bitcoin in these cases. Therefore, the collected results are the best that can be obtained by using the given survey method.

5. Socio-Economic Factors Behind Bitcoin Use Cases

Different socio-economic factors can motivate people to use Bitcoins. In this section we propose a link between the main socio-economic drivers and bitcoin use cases discovered in the given study. We introduce the following factors:

(1) Profit generation.
(2) Weak economy of the region, including low GDP, high inflation, extractive economic institutions, high level of unemployment, etc.
(3) Illicit activity, including money laundering, terrorism financing, drug dealing, etc.
(4) Entertainment.

Based on the above-mentioned factors, we divide the bitcoin use cases into the subsequent categories:

Profit generation	Weak regional economy	Illicit activity	Entertainment
Trading			
Investing	Investing		
	Remittance		
	Salary	Salary	
			Gambling
			Learning
		Payment method	
	Store of value		

In order to provide an illustration of this hypothesis, the given sample was divided into 5 regions: Latin America, Africa, Central Europe, Eastern

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Europe and Asia. Each region includes countries that have the highest users' representation on the LocalBitcoins platform, respectively.[2]

As stated above, the majority of users tend to use Bitcoin for trading and investment purposes, however, some regional specifics can be detected. Users from Latin America demonstrate the highest numbers in Bitcoin usage as a store of value and remittance in relation to other regions. Besides using Bitcoin as a tool for investments, Central and East Europe display the highest numbers in Bitcoin usage as a payment method. In the case of users coming from African countries, the absolute majority tend to use Bitcoin for profit generation through trading and investments. From the remaining regions, the African region shows a higher interest towards learning purposes. Finally, the Asian region exhibits strong usage of Bitcoin for trading and investment benefits.

Taking into account the classification of developing and developed countries provided by the United Nations (2021), developing economies represent higher interest in trading, store of value and remittance than developed ones. On the other hand, countries of Central Europe tend to use Bitcoin for profit generation through the medium of investments.

6. Conclusion

In this study, we aimed to understand bitcoin usage worldwide by analyzing and classifying the major bitcoin use cases to better understand what drives its users to use bitcoin. It is important to emphasize that the majority of related studies have employed analysis of public record of bitcoin activity, therefore research results are heuristic in nature and sometimes lack direct access to personal metadata linked to transactions.

Moreover, due to the recent worldwide adoption of bitcoin, the structure of bitcoin users has been replenished and is constantly changing. Thus, an updated understanding of the structure of current bitcoin users' is required.

Besides the fact that some professionals tend to consider bitcoin as a bubble (Krugman, 2018), or believe that it does not serve any social function (Hochstein, 2017), the results of the research conducted demonstrate that the scope of Bitcoin usage is noticeably diverse and not merely limited to illegal usage as often claimed and construed. More specifically, the largest Bitcoin use cases are trading (25.5% of all respondents), investing (24%), using Bitcoin as a payment method (13.9%), store of value (10.4%) and finally remittance (9.7%).

[2] Corresponding data can be found under the Data Appendix section.

LocalBitcoins is dedicated to further investigating global use cases in the future too, in order to trace the dynamics and shifts of bitcoin use cases.

Acknowledgment

We would like to thank the LocalBitcoins employees and organization for their hard work.

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Data Appendix

Country	Trading	Investments	Payment method	Store of value	Remittance	Learning	Gambling	Salary	Other
Latin America									
Columbia	36	26	7	6	11	8	1	2	3
Argentina	16.2	26	4	17.9	20.3	8	2	2	3.2
Venezuela	23.6	16.8	8.7	24	10.3	9	0.7	3.5	4.7
Brazil	12.3	23.9	9.7	5.6	10.3	27.0	1	1	8
Peru	24.1	28.5	9	6.1	16.1	10	1.1	1.2	3.9
Africa									
Kenya	35.1	30.2	8.8	4.3	8.1	9.6	1	1	1.5
Nigeria	34.7	28.4	5.8	4.6	4.2	17.7	—	—	4.2
South Africa	39.7	27.7	11.7	4.3	5	6.9	1.3	2.1	1.3
Ghana	23	29.2	12	5.2	6.7	18.1	0.6	0.9	4
Central Europe									
United Kingdom	18.7	32.8	23.7	6.5	4.4	6.9	2.6	0.9	3.6
Sweden	16.7	24.4	30	6.2	1.1	11.8	—	0.8	2.1
Spain	23	22.8	6.1	7.7	26.6	10.2	0.8	1	1.8
France	28.5	28.5	24.9	5.9	5	8.6	4.5	1.4	3.6
Germany	19.4	26.9	24.7	4.4	8.8	9.3	1.3	3.1	2.2
Eastern Europe									
Russia	19.1	20.9	28.1	9.3	4.2	9.8	3.8	2.2	2.6
Belarus	19.9	17.8	31.6	10.4	3	8	6.3	1.7	1.3
Ukraine	25.1	26.6	15	11.7	4.8	11.2	2	1.5	2
Asia									
India	40	31.3	10.7	3	3.7	13.3	—	—	3.7
China	23.2	37.1	15.3	9.1	8.2	5	0.5	0.9	0.7
Philippines	34.8	34.8	9.6	1.5	5.9	6.7	0.7	3	3

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